Multi-Object Tracking using YOLOv8 and DeepSORT

1. Introduction

In this project, a multi-object tracking system is developed to detect and track multiple people in a video. The main objective is to assign a unique ID to each detected person and maintain that identity across frames as the video progresses.

This type of system is widely used in applications such as surveillance, traffic monitoring, and sports analysis.

2. Problem Statement

The goal of this assignment is to build a system that can:

- Detect multiple objects (people) in a video
- Track each object across frames
- Assign a unique ID to each object
- Maintain ID consistency even in challenging scenarios like movement, occlusion, and overlap
- Generate an output video with bounding boxes and IDs

3. Approach

The solution is based on a tracking-by-detection pipeline, which consists of two main stages:

1) Object Detection

For object detection, YOLOv8 is used. It is a real-time object detection model that processes each frame independently and identifies objects along with their bounding boxes.

In this project, only the "person" class is considered for detection.

2) Object Tracking

For tracking, DeepSORT is used. It extends the SORT algorithm by combining motion and appearance information.

- A Kalman Filter is used to predict object positions in the next frame
- The Hungarian Algorithm is used to match detections with existing tracks

- Appearance features are used to distinguish between similar objects

Each object is assigned a unique track ID, which is maintained as long as the object is tracked successfully.

4. Implementation Details

- Programming Language: Python
- Libraries Used:
    - OpenCV for video processing
    - NumPy for numerical operations
    - YOLOv8 (Ultralytics) for detection
    - DeepSORT for tracking

The system processes the input video frame by frame. For each frame:

1. Objects are detected using YOLOv8
2. Detections are passed to DeepSORT
3. Tracker assigns IDs and updates object positions
4. Bounding boxes, IDs, and trajectories are drawn
5. Frame is saved to output video

5. Output

The output is an annotated video that includes:

- Bounding boxes around each detected person
- Unique IDs assigned to each person
- Current number of people visible in the frame
- Total number of unique IDs generated
- Trajectory lines showing movement

6. Challenges Faced

- ID switching when objects are occluded or leave and re-enter the frame
- Handling fast movement and overlapping objects

- Maintaining stable tracking in crowded scenes

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7. Limitations

- The system may assign a new ID if a person disappears and reappears
- Accuracy depends on video quality and lighting conditions
- Only the "person" class is considered

8. Future Improvements

- Implement entry/exit-based counting
- Improve tracking using stronger re-identification models
- Support multiple object classes
- Add features like speed estimation and heatmaps

9. Conclusion

This project successfully demonstrates a multi-object tracking system using YOLOv8 and DeepSORT. The system is able to detect and track multiple people in a video while maintaining unique identities across frames.

It provides a good foundation for building more advanced tracking systems for real-world applications.

10. References

- YOLOv8 Documentation (Ultralytics)
- DeepSORT Algorithm
- OpenCV Documentation